CHAPTER-5: Implementation

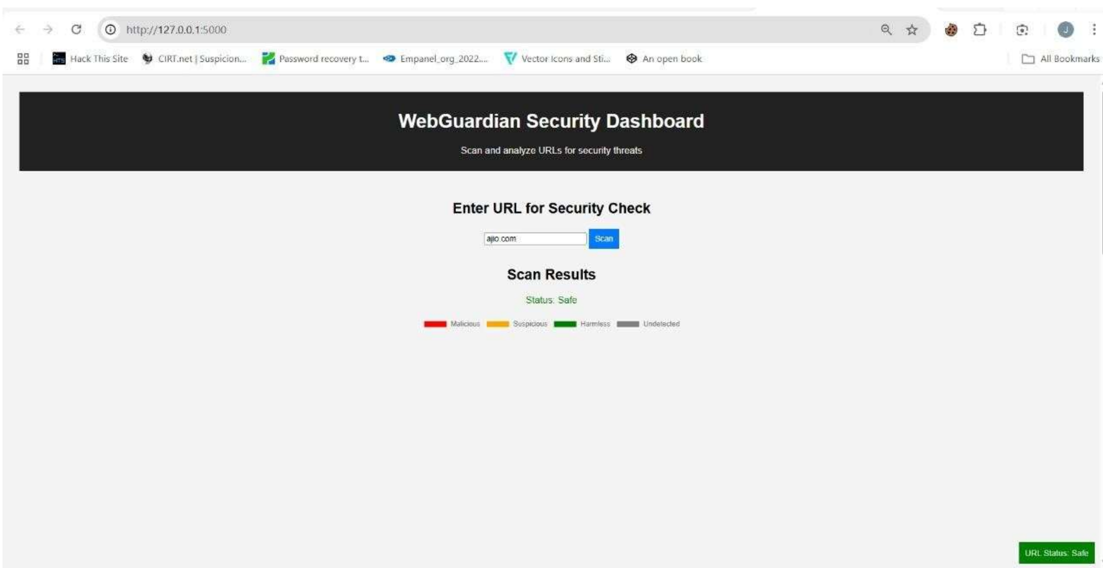
5.1 Overview

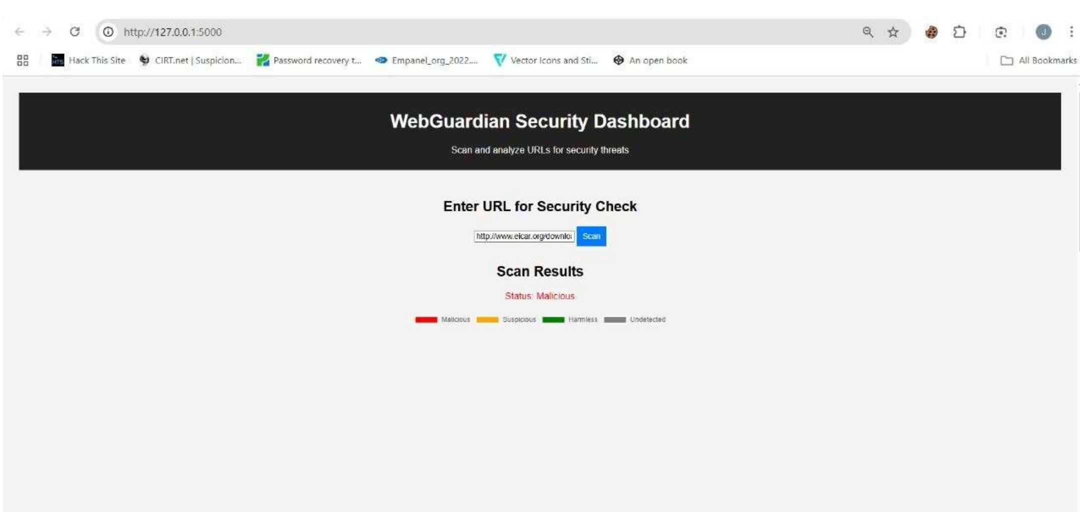
This module helps detect malicious URLs and domains. Currently, we use VirusTotal API to check URLs, and in the future, we will integrate Machine Learning for improved detection.

5.2 Current Implementation (Using VirusTotal API)

5.2.1 How It Works

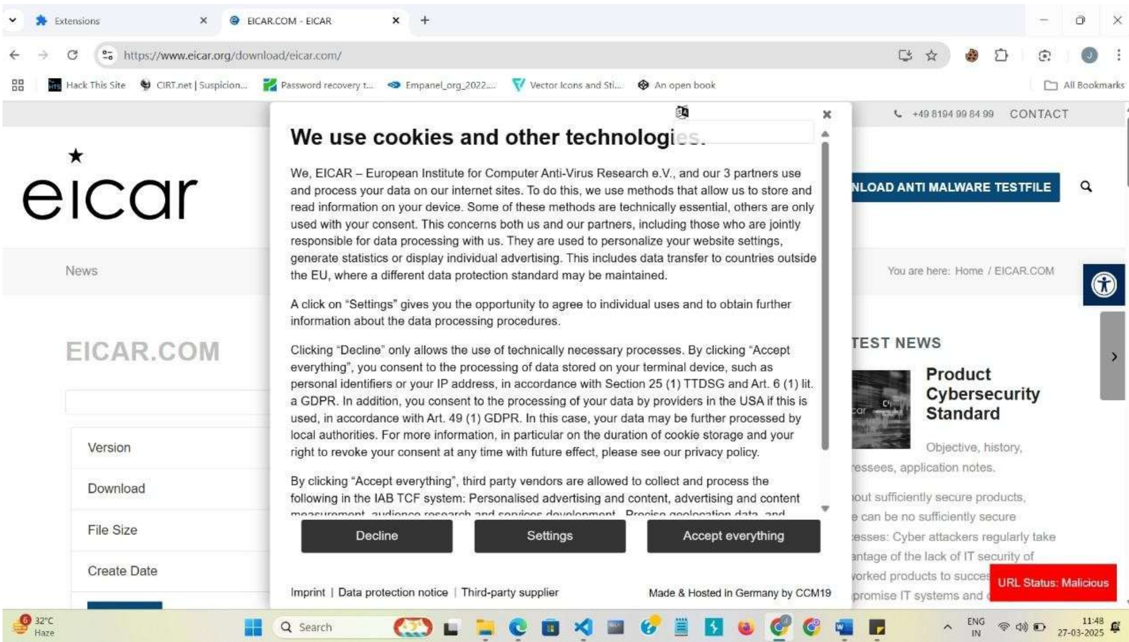
1. **User enters a URL or domain.**
2. **The system sends it to VirusTotal API.**
3. **VirusTotal returns a security report.**
4. **Results are displayed to the user.**

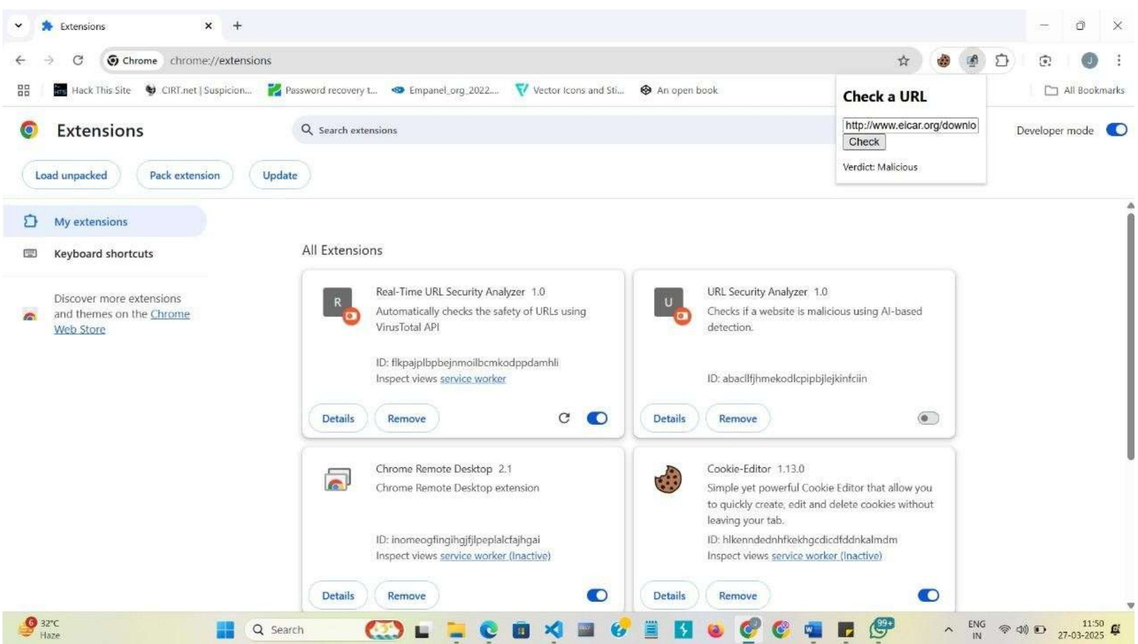




5.2.2 Steps We Implemented

- ✓ **Extract domain from URL**
- ✓ **Send request to VirusTotal API**
- ✓ **Retrieve scan results**
- ✓ **Display results in frontend**

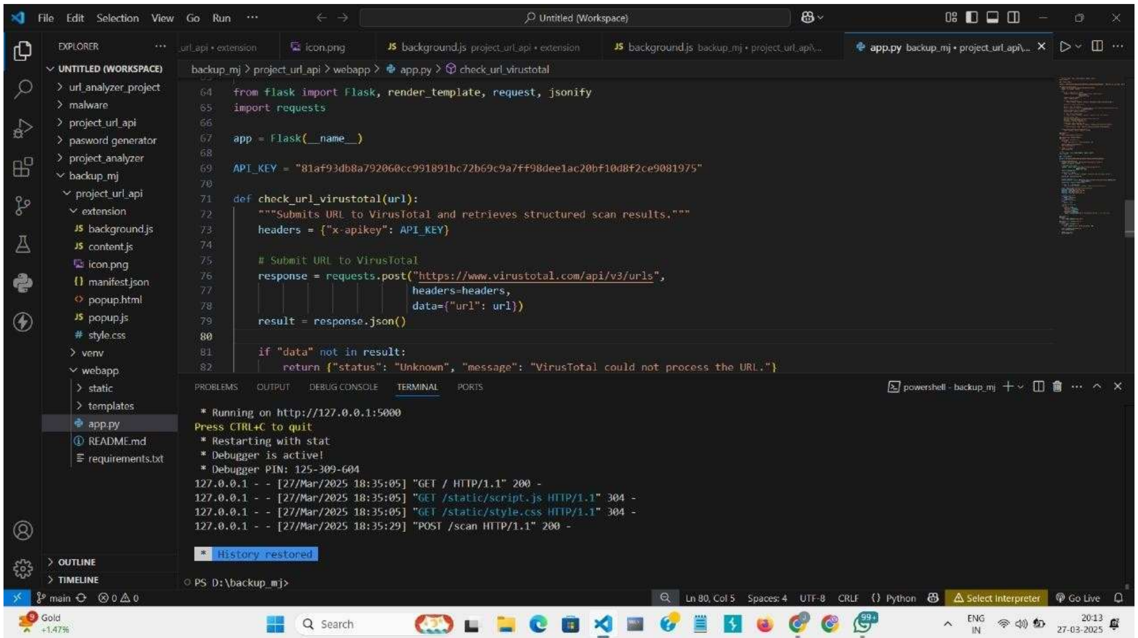


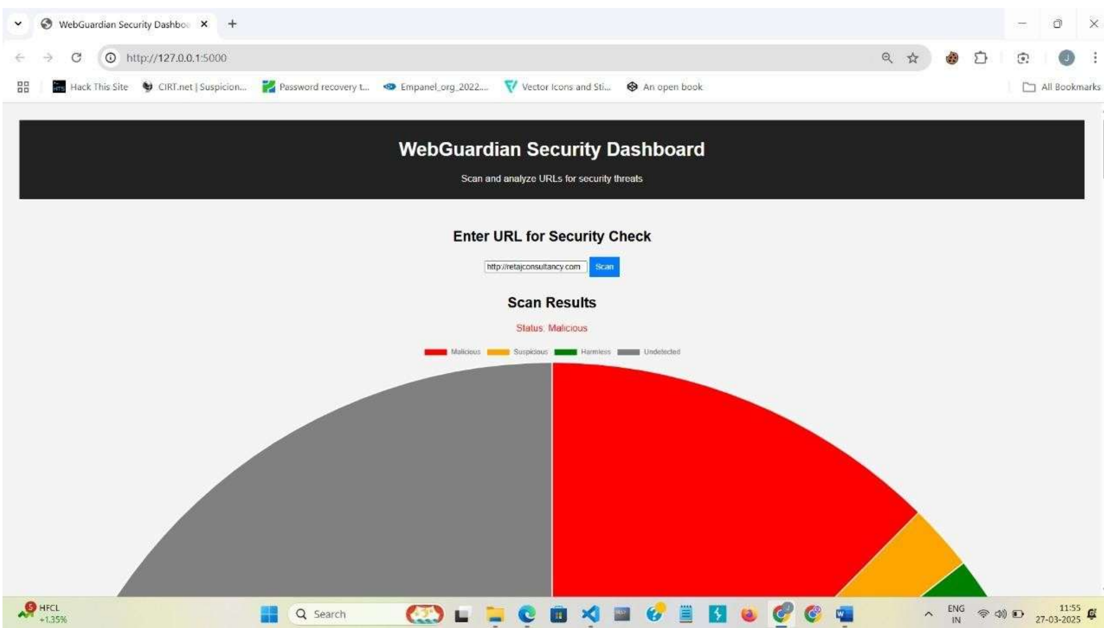


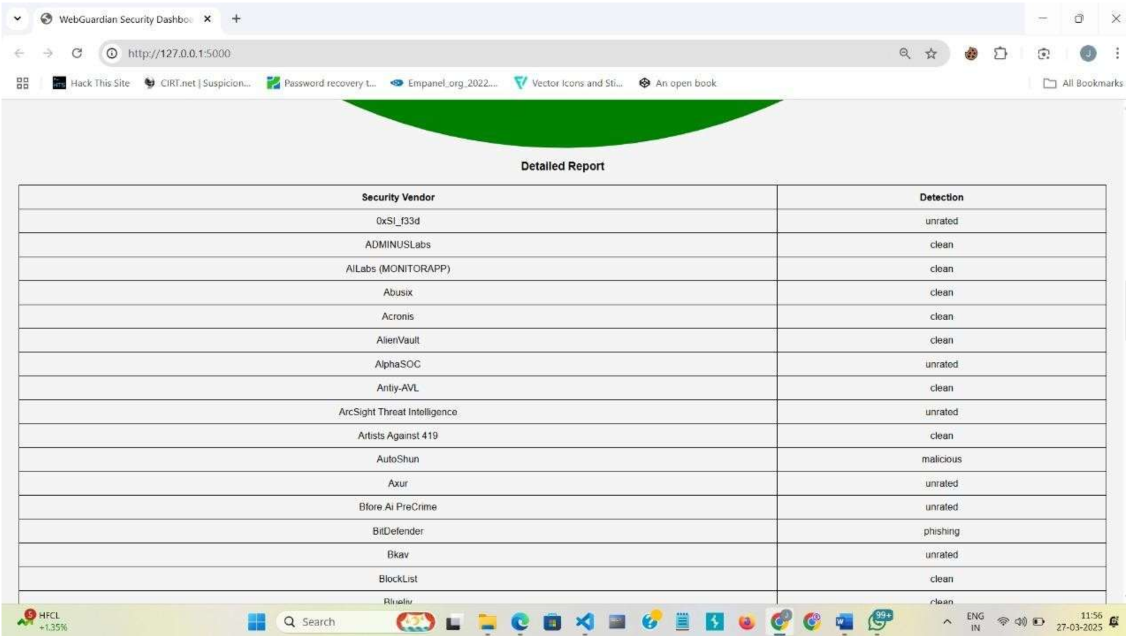
5.2.3 Code Implementation

1. **User enters URL/domain in the frontend.**
2. **Backend (Flask) sends request to VirusTotal API.**

3. **Response is parsed, and the result is displayed.**







Security Vendor	Detection
0xSI_f33d	unrated
ADMINUSLabs	clean
AILabs (MONITORAPP)	clean
Abusix	clean
Acronis	clean
AlienVault	clean
AlphaSOC	unrated
Antiy-AVL	clean
ArcSight Threat Intelligence	unrated
Artists Against 419	clean
AutoShun	malicious
Axur	unrated
Bfore Ai PreCrime	unrated
BitDefender	phishing
Bkav	unrated
BlockList	clean

5.2.4 Conclusion

- ✓ **Current Implementation – VirusTotal API for scanning URLs.**
- ✓ **Future Enhancement – Machine Learning for detecting new threats.**